

Satida Sookpong

Intro to NLP

Foundations of Natural Language Processing

1323326 การประมวลผลภาษาธรรมชาติ (Natural Language Processing)
1-0303 Computer & Sound; Lab 2

Outline

- **What is NLP**
- **History**
- **NLP Tasks**
- **Main component**
- **Challenge**
- **Application**
- **Tools**

What?

What is NLP??



“NLP is the science that integrates linguistics, computer science, and artificial intelligence to enable computers to understand, interpret, and generate human language.”

Ref. Huaping_

Human Language (Meaning) <-> computer language

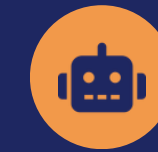
Built from 3 fields



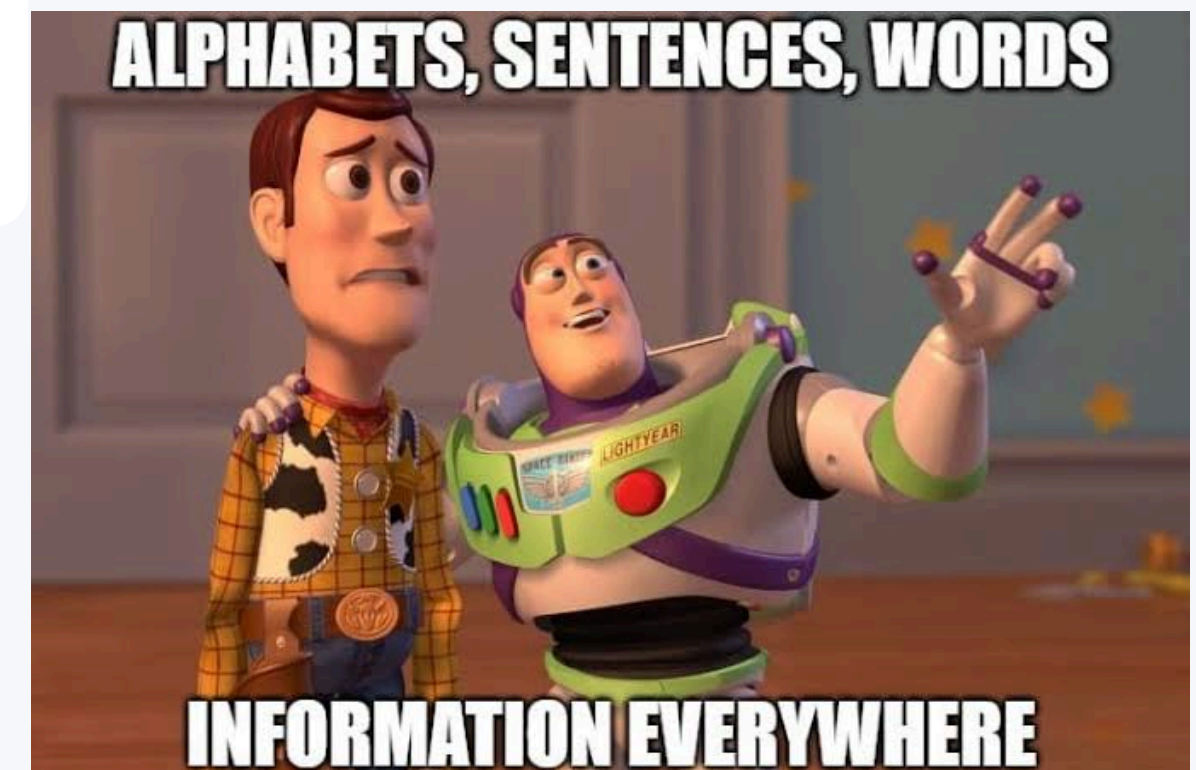
Linguistics



Computer Science



Artificial Intelligence



History

A Brief History of NLP

1

Before 1960s

Machine
Translation
(rule-based)

2

1960s–70s

Early AI
on NLP

3

1970s–80s

Grammatical
Logic

4

1980s–2000s

AI & Machine
Learning

5

2010s–2017

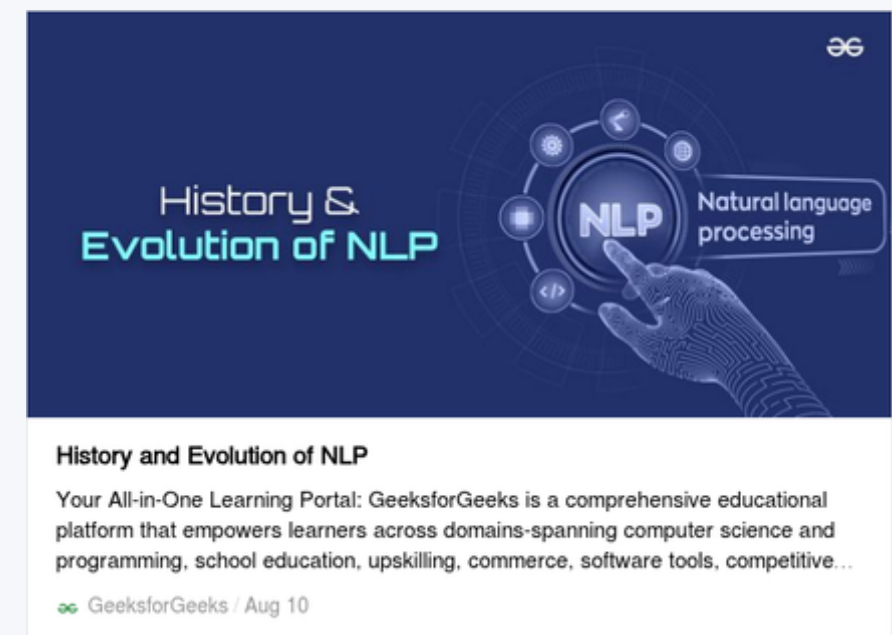
Big Data &
Deep Networks

6

2017–2026

Transformers
& LLMs

philosophy → rules → statistics → learned representations → massively pretrained transformers.

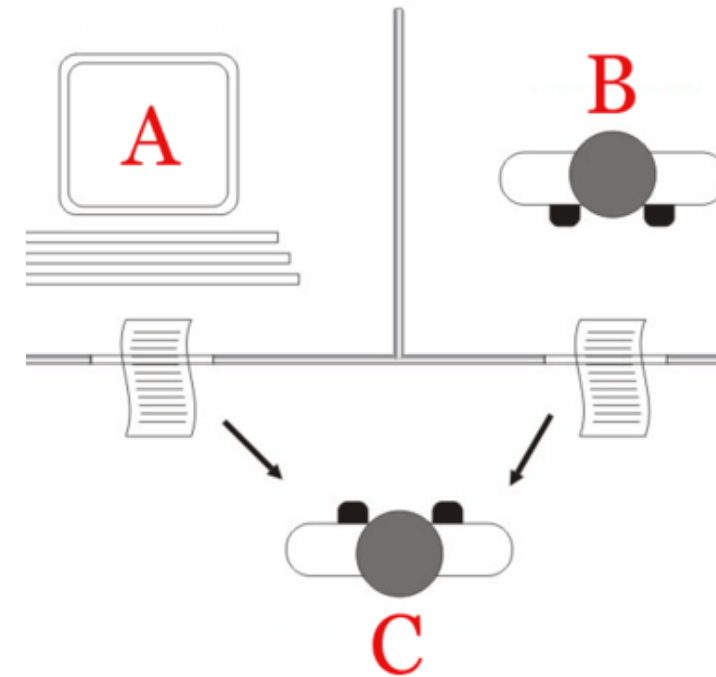


Machine Translation — Before the 1960s



Philosophical Roots

Studied the relationship between words and languages



- 1950 — Alan Turing publishes "**Computing Machinery and Intelligence**" and proposes the **Turing Test**
- 1952— International Conferences on Machine Translation use **rule-based** and **stochastic techniques**
- 1957 — Noam Chomsky introduces universal grammar: **Chomsky Hierarchy**
 - Human define knowledge

Early AI on NLP — 1960s to 1970s



1961 — BASEBALL System

Green et al: **Q&A-based domain expert system**

- Input: only one domain in key-value
eg. {Month: July, Place: Boston, Team: Red Sox, Score: 5}



1968 — Minsky's Inference Engine

Marvin Minsky: **AI-based question-answering system**

- knowledge-based **interpretations** of questions and answers.



1970 — Augmented Transition Networks

William A. Woods: **ATNs to represent natural-language input**

- encode real-world knowledge -> machine-understandable form.

Grammatical Logic — 1970s to 1980s



Knowledge, Logic & Reasoning

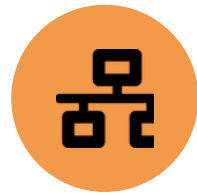
"grammatical logic"

- Powerful sentence-processing
- early Q&A chatbots
- SRI's Core Language Engine (CLE)

text (message) → morphology (โครงสร้างคำ) → syntax → semantics → logical form → context (บริบท) → meaning

- Discourse Representation Theory (DRT) - ศึกษาความหมายข้ามประโยค

AI and Machine Learning — 1980s to 2000s



The Hopfield Network

Prof. Emeritus John Hopfield

Hopfield Network

- using **machine learning** - complex **rule-based** and **stochastic methods**.
- Hand crafted feature



Corpus Linguistics & Lexicalization

Upgrades in computational power and memory

- **Chomsky's linguistic theory**
corpus-based machine learning
- **"NLP lexical / corpus"** -
lexicized grammar



2006 — IBM DeepQA Project

Dr. David Ferrucci, IBM

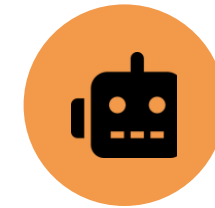
- **DeepQA** - question-answering system
- Large algorithm - learn from data

AI, Big Data & Deep Networks — 2010s Onward



Cloud, Mobile & Big Data

- Cloud-computing, Mobile computing
- big data fuel, deep-network analysis
- Recurrent neural networks (RNNs) using Long Short-Term Memory (LSTM) and related architectures become the workhorse for sequence modeling.



Big Tech Drives the Field

Google, Amazon, and Facebook invest heavily in agent technologies and deep neural networks, developing:

- (2013 - Google) Word2Vec: word as a vector
- (2014): Seq2Seq, Attention
- (2016) Google NMT: Google translate

Transformers & the Modern LLM Era — 2017–2026



2017

The Transformer

“Attention is all you need” - Vaswani et al., Google (2017)

- self-attention
- Transformer

arxiv.org/abs/1706.03762



2018–2020

Pretrained Giants

- BERT, GPT-2, GPT-3
- pretraining one giant model → adapting it

arxiv.org/abs/1706.03762



Nov 2022

ChatGPT

- LLMs
- OpenAI’s chat interface reaches **1 million users** in five days

arxiv.org/abs/2005.14165



2023–2026

The Frontier Race

- GPT-4/5, Claude, Gemini, LLaMA, DeepSeek: multimodal
- reasoning, coding and agentic capability

NLP today

Wayne Xin Zhao et al. A Survey of Large Language Models

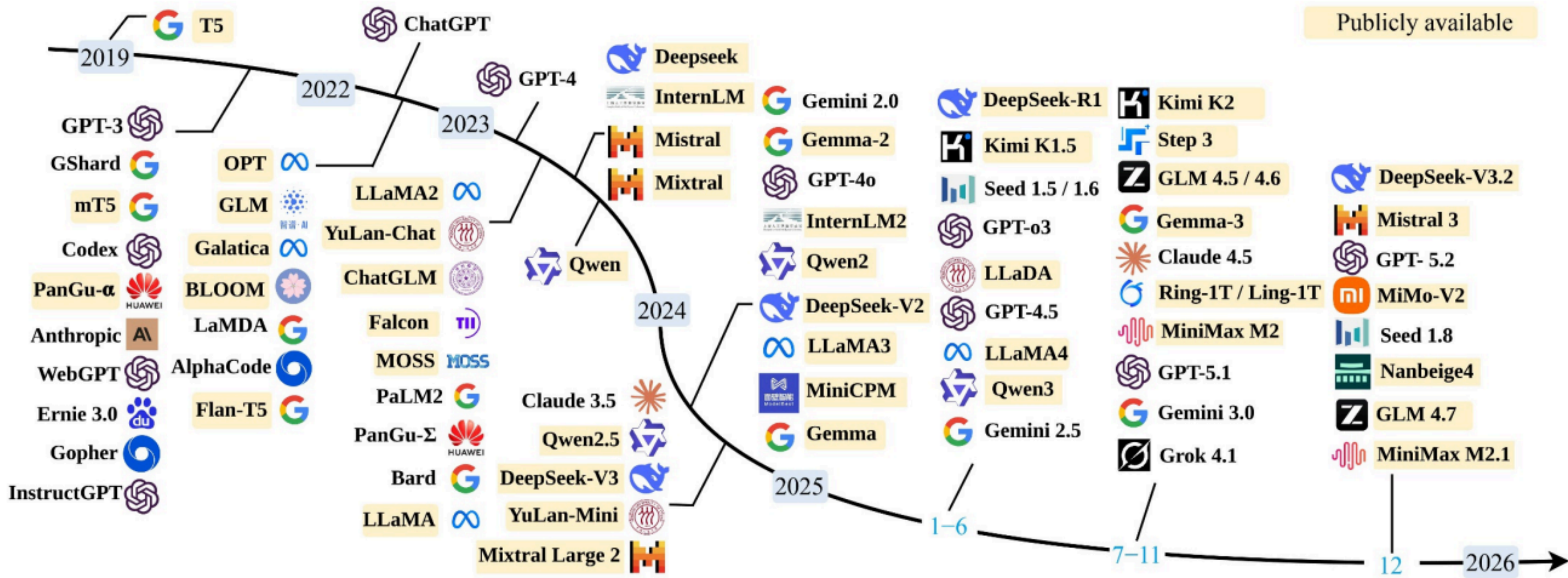


Fig. 1 A timeline of representative LLMs released in recent years. Models with publicly available checkpoints are highlighted in yellow

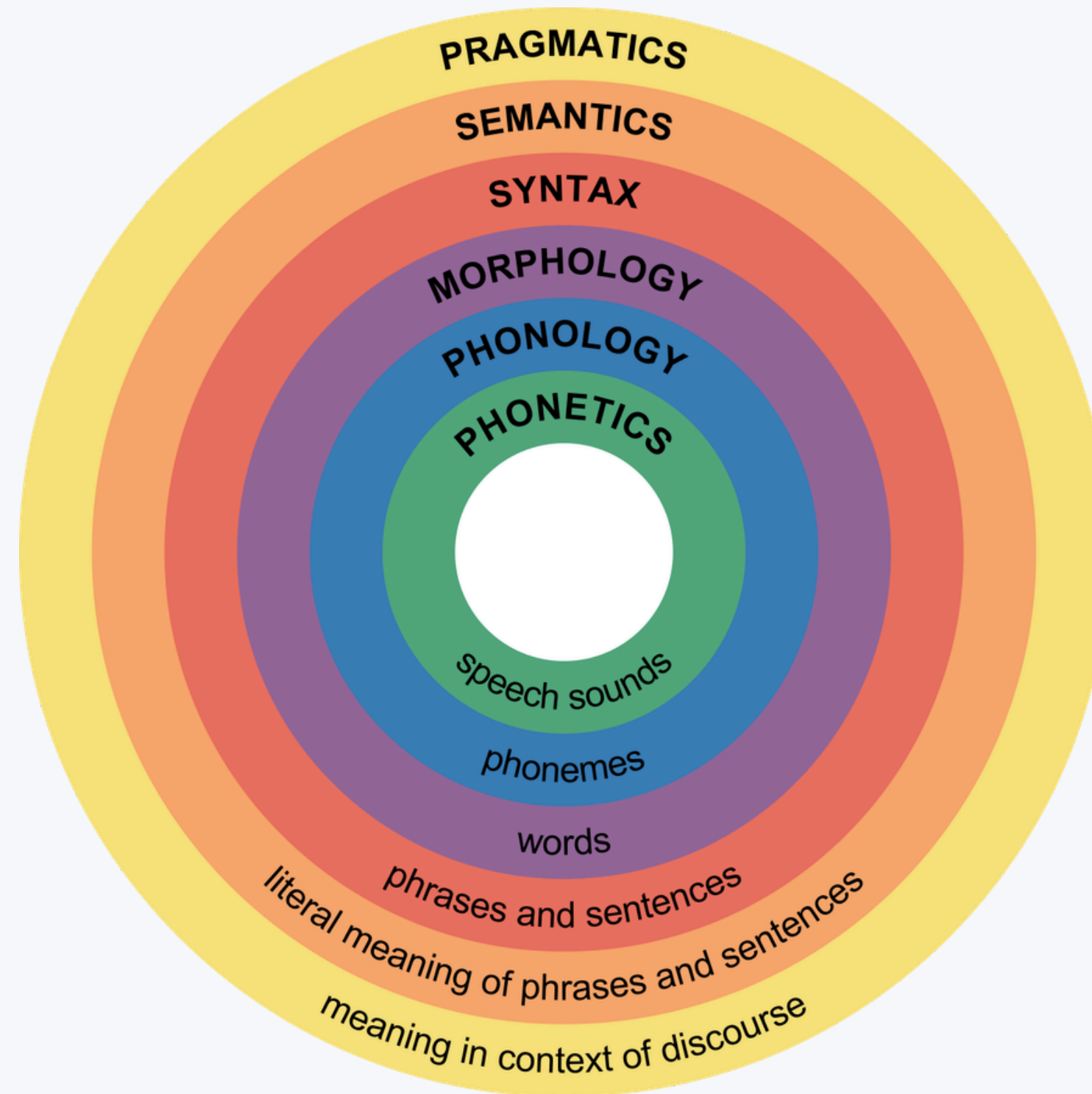
Benchmark

llm-stats.com

RANK	MODEL	LLM STATS ? ↓	REASONING ?	CODING ?	AGENT ?	CONTEXT ?	SPEED ?	PRICING \$/M ?
1	 GPT-6 Astra ★ – NEW OpenAI	60.7	58.7	49.0	46.4	1.1M	118 c/s	\$11.90
2	 Claude Fable 5.1 – NEW Anthropic	56.8	53.9	36.0	41.9	1.0M	49 c/s	\$11.90
3	 Claude Opus 5 ↓ 1 Anthropic	55.4	54.3	42.3	39.9	1.0M	133 c/s	\$5.95
4	 GPT-5.6 Sol ↓ 3 OpenAI	55.3	54.4	46.0	41.0	1.1M	79 c/s	\$6.19
5	 Muse Spark 1.3 ★ – NEW Meta	55.3	52.8	41.7	40.4	1.0M	89 c/s	\$0.10
6	 Claude Mythos Preview – UNRELEASED Anthropic	55.1	55.9	44.9	36.2	—	—	—
7	 Claude Fable 5 ↓ 4 Anthropic	54.8	52.1	45.9	40.3	1.0M	138 c/s	\$11.90
8	 Kimi K3 ↓ 4 Moonshot AI	53.8	52.6	43.4	39.1	1.0M	32 c/s	\$3.57

NLP task

Levels of languages



Levels of languages



The 5 Steps in Natural Language Processing (NLP)

1. Lexical analysis

2. Syntactic analysis

3. Semantic analysis

4. Discourse integration

5. Pragmatic analysis

1. Text → Paragraph Sentence Word
2. Grammar + relationship between word
3. Exact meaning (one sentence)
4. Meaning of the sentence (based on previous sentence - pronoun)
5. Actual meaning based on context - real world knowledge

Upstream & Downstream Tasks

UPSTREAM TASKS

Linguistic preprocessing

Pre-training model

- tokenization
- tagging
- parsing.

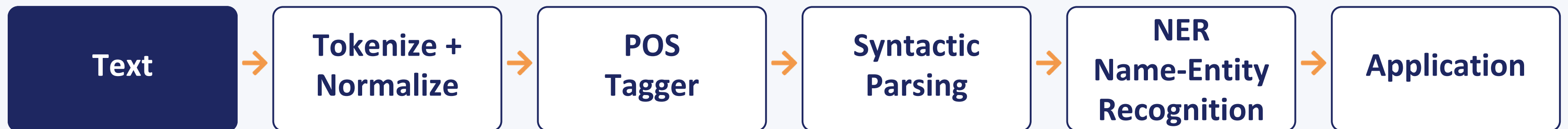
DOWNSTREAM TASKS

Applied tasks built on upstream output

Specific Application

Fine-tuning

- classification
- translation
- Q&A.



Main component

Main Components of NLP



NLU

Natural Language Understanding

Interprets meaning, intent, and structure from human input.



KAI

Knowledge Acquisition and Inferencing

Applies knowledge representation and inference (reasoning) to process intent.



NLG

Natural Language Generation

Produces fluent, human-readable text or speech as output.



Challenge

Why NLP is Hard: Core Challenges



Abstractness

Meaning is rarely literal; words encode concepts beyond their surface form.



Combinability

Words combine in near-infinite ways, producing new, unseen meanings.



Evolutiveness

Language constantly evolves with new slang, terms, and usage.



Nonstandardness

Typos, dialects, and informal grammar deviate from clean rules.



Subjectivity

Tone, sentiment, and intent vary by speaker, context, and culture.

“ ภาษาไทยง่ายเพราะเราเป็นคนไทย ”

Why NLP is Hard: Core Challenges



Abstractness

ความเป็นนามธรรม

Meaning is rarely literal; words encode concepts beyond their surface form.

ไปให้แม่แก่ทำ
สวย → สวยที่มอง = ไม่สวย
ได้ → ได้อยู่ = ไม่ได้

Why NLP is Hard: Core Challenges



Combinability

ผสมกันได้

Words combine in near-infinite ways, producing new, unseen meanings.

แม่
แม่น้ำ
แม่ยก
แม่ไม้มวยไทย

กับ = with
ข้าว = rice
กับข้าว = no rice

When you learn Thai:

Thai By
Chom



ไป (bpai)
= to go



สบาย (sà-baai)
= comfortable



ไปสบาย
(bpai sà-baai)
= passed away

Why NLP is Hard: Core Challenges

ใช้ไหม → ใช้ปะ → ชะ
เลื๑ศ → เลื๑ด
จะรื๑ว



Evolutiveness

มีวิวัฒนาการ

Language constantly evolves
with new slang, terms, and
usage.

Why NLP is Hard: Core Challenges

กำลัง จะ ไป กินข้าว อยู่ แล้ว

- กำลังจะไปกินข้าว
- กินข้าวอยู่
- กินข้าวแล้ว



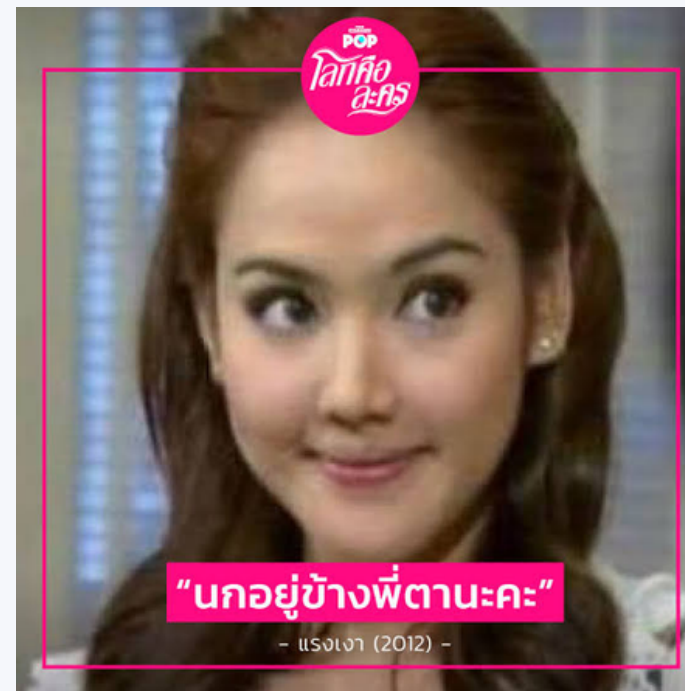
Nonstandardness

ไม่มีหลักการที่แน่นอน

Typos, dialects, and informal grammar deviate from clean rules.

Why NLP is Hard: Core Challenges

แกงส้ม - แกงเหลือง
“ไปส่ง” กินข้าวหน่อย

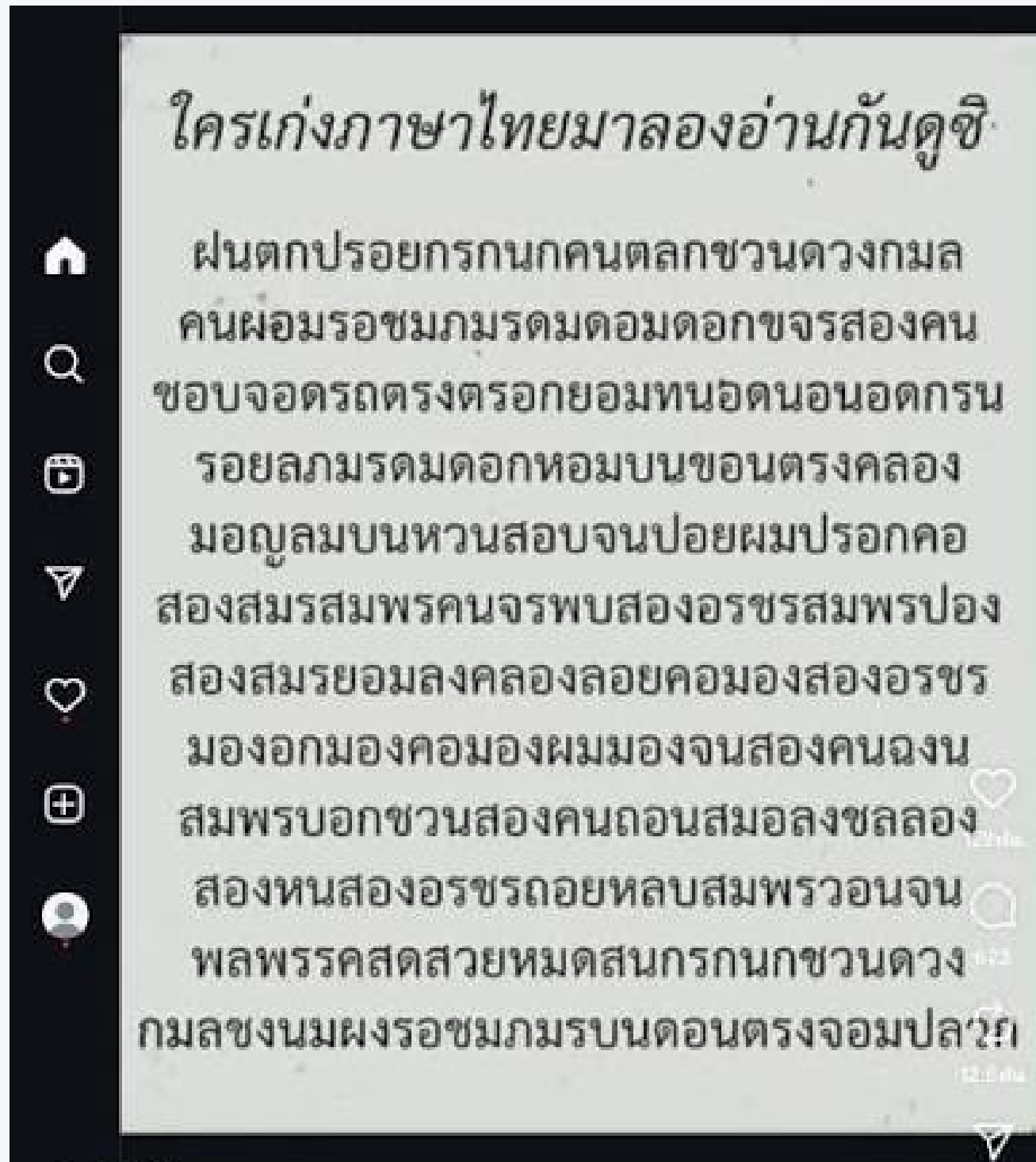


Subjectivity

ขึ้นกับแต่ละบุคคล/บริบท

Tone, sentiment, and intent
vary by speaker, context, and
culture.

Why NLP is Hard: THAI NLP Challenges



คุณอ่านภาษาไทยเก่งแค่ไหน? ลองดู
..หมาหันมา หาหมี เห็นหุหมา
หุหมาหนา หมีหมี หมีเห็นหมี
เห็นหมีหัน หุหมี หมีหันดู
หมีหมาหา เห็นหมี หันหุดี
หาหมีเห็น เหม็นหุ หมีหุเหม็น
หาหมีหาย หมาเห็น เหม็นหุหมี
หมาหมีหาย หมาหมี หุหมี
หมาหาหมี หมาหมี หมีหายตัว



Segmentation

การตัดคำภาษาไทย

word segmentation
sentence segmentation



Application

APPLICATIONS

NLP Applications



Text Classification

Assigning labels like spam or topic categories.



Sentiment Analysis

Detecting positive, negative, or neutral emotion.



Machine Translation

Converting text from one language to another.



Named Entity Recognition

Identifying people, places, dates, etc.



Text Summarization

Shortening text while keeping key meaning.



Question Answering

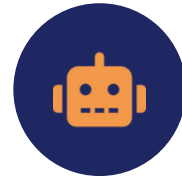
Reading text to return exact answers.

NLP Applications



Voice Assistants

Alexa, Google Assistant, Siri



Customer Chatbots

Automated support
conversations



Spam Filtering

Email classification &
security



Predictive Typing

Auto-correct & text
suggestions



Translation Tools

Cross-language
communication



Sentiment Tracking

Social media monitoring



Document Search

Search & recommendation
systems



Tools

Popular NLP Libraries

NLTK

Natural Language Toolkit — the classic Python library for teaching and research; rich in classic algorithms and corpora.

spaCy

Industrial-strength NLP — fast, production-ready pipelines for tokenization, tagging, parsing, and NER.

PyThaiNLP

Thai Natural Language Processing — open-source toolkit purpose-built for Thai language tasks.